6.4 Exponential Functions

Name _____ Period ____ Date _____

Algebra I 6.4 (30-minute plan) · Student sheet

I can…

- identify and evaluate exponential functions.
- graph exponential functions and find their asymptote, domain, and range.
- tell whether a table or situation is linear or exponential.

Vocabulary · fill in the missing word

Exponential function — y = _____, where a ≠ 0, b ≠ 1, and b > 0

Asymptote — a line a graph gets closer and closer to but never _____

Linear vs. exponential — linear: add the same _____ each step; exponential: multiply by the same amount

Part 1 · Notes and examples (fill in each note, then work the example)

Exponential function: An **exponential function** has the form y = _____, where a ≠ 0, b > 0, and b ≠ 1.

Example 1

Does each equation represent an exponential function?

a. $y = 4(7)^x$ b. $y = -6x$ c. $y = x^3$

Linear or exponential tables: Linear tables have a constant _____. Exponential tables have a constant _____.

Example 2

Is each table linear or exponential?

a. x: 0, 1, 2, 3 y: 3, 6, 12, 24

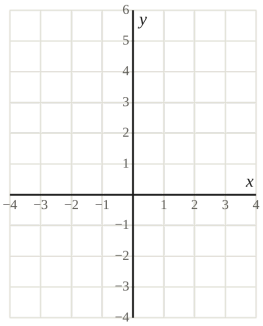
b. x: 0, 1, 2, 3 y: 1, 4, 7, 10

Graphs: The graph of $y = ab^x + k$ has an asymptote at _____. The domain is _____.

Exponential situations: A situation is exponential when it changes by the same _____ each time.

Example 3

Graph $y = 2^x$ and $y = 2^x - 3$. Find the y-intercept, asymptote, domain, and range of $y = 2^x - 3$.



Part 2 · Partner practice

Partner practice

Evaluate $y = 4(3)^x$ when $x = 2$.